

STAT 154 Notes

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Least Squares

- We consider the following optimization problem:

$$\min_{\theta \in \mathbb{R}^d} \frac{1}{2} \|\mathbf{X}\theta - \mathbf{y}\|_2^2 \quad (\text{Least Squares})$$

where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the feature matrix and $\mathbf{y} \in \mathbb{R}^n$ is the set of observations.

- Note that the normalization factor $\frac{1}{2}$ is used for convenience later on—it does not change the optimization problem.
- When $n \geq d$ and $\mathbf{X}$ is full column rank, one way to compute the solution to this problem is using the closed form expression:

full column rank
rank(X)=d
mean no perfect multicollinearity

$$\theta^{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Convex optimization

- We can also study least squares in the framing of convex optimization.
- Consider the following optimization problem

$$\min_{\theta} f(\theta) \quad \text{where } f \text{ is convex.} \quad (1)$$

- Note that because of convexity all minimizers θ^* satisfy

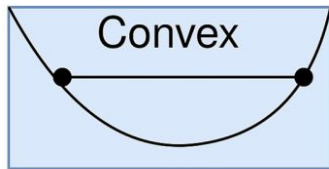
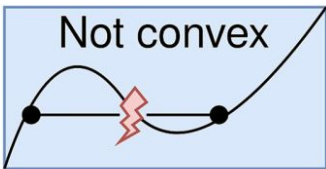
$$\nabla_{\theta} f(\theta^*) = 0.$$

In fact the closed form for θ^{OLS} is obtained by solving for this equation.

*Left side: The value of the function at a point between b and c.
Right side: The value on the straight line connecting the two points.*

lambda: [0,1]

$$f[\lambda b + (1 - \lambda)c] \leq \lambda f(b) + (1 - \lambda)f(c)$$



Gradient Descent

- A popular algorithm for finding these stationary points is gradient descent

$$\theta^{t+1} = \theta^t - \eta \nabla_{\theta} f(\theta^t).$$

- Note that these updates will not move when $\theta^t = \theta^*$.
- Note that the direction of $-\nabla_{\theta} f$ is also the direction of steepest descent so for small enough η , the gradient step will reduce the function value.
- We now see some details of gradient descent for least-squares.

Gradient Descent on Least Squares

- We now study how the gradient descent method behaves when applied to the Least squares problem.
- First we expand the objective so that gradient computation is easy:

$$f(\theta) = \frac{1}{2} \|\mathbf{X}\theta - \mathbf{y}\|_2^2 = \frac{1}{2} \theta^\top (\mathbf{X}^\top \mathbf{X}) \theta + \frac{1}{2} \mathbf{y}^\top \mathbf{y} - \theta^\top \mathbf{X}^\top \mathbf{y}.$$

- Then we have

$$\nabla_{\theta} f(\theta) = \mathbf{X}^\top \mathbf{X} \theta - \mathbf{X}^\top \mathbf{y}.$$

And hence gradient descent updates with step size η become:

$$\begin{aligned} \theta^{t+1} &= \theta^t - \eta \nabla_{\theta} f(\theta^t) \\ &= \theta^t - \eta (\mathbf{X}^\top \mathbf{X} \theta^t - \mathbf{X}^\top \mathbf{y}). \end{aligned}$$

How do we choose the step size?

Gradient Descent on Least Squares

- We now derive the recursion in the error. To simplify our calculations, we assume that there exists θ^* such that $\mathbf{y} = \mathbf{X}\theta^*$.
- Under that assumption, the gradient descent updates simplify to

$$\begin{aligned}\theta^{t+1} &= \theta^t - \eta((\mathbf{X}^\top \mathbf{X})\theta^t - \mathbf{X}^\top \mathbf{y}) \\ &= \theta^t - \eta((\mathbf{X}^\top \mathbf{X})\theta^t - \mathbf{X}^\top \mathbf{X}\theta^*) \\ &= \theta^t - \eta \mathbf{X}^\top \mathbf{X}(\theta^t - \theta^*).\end{aligned}$$

- As a result, we have

theta* is the optimal solution

$$\begin{aligned}\theta^{t+1} - \theta^* &= \theta^t - \eta \mathbf{X}^\top \mathbf{X}(\theta^t - \theta^*) - \theta^* \\ &= (\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})(\theta^t - \theta^*)\end{aligned}$$

expanding this term t times

error after t steps

$$= \underbrace{(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^{t+1}}_{\text{convergence controlled by this term}}(\theta^0 - \theta^*).$$

convergence controlled by this term

Gradient Descent on Least Squares

To understand how fast the error shrinks, we analyze its operator norm (maximum factor by which it lengthens vectors)

- Now to choose the step size, we see that

$$\begin{aligned}\|\theta^t - \theta^*\|_2 &= \|(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})^t (\theta^0 - \theta^*)\|_2 \\ &= \|(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})\|_2^t \|\theta^0 - \theta^*\|_2\end{aligned}$$

where we use $\|\mathbf{A}\|_2$ to denote the operator norm of the matrix.

- If the eigenvalues of the matrix $\underbrace{\mathbf{X}^\top \mathbf{X}}_{\text{symmetric}}$ lie between m and L , then we have

$$\lambda(\underbrace{\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}}_{\text{symmetric}}) \in [1 - \eta L, 1 - \eta m].$$

Gradient Descent on Least Squares

Since $(\mathbf{X}^\top \mathbf{X})$ is symmetric, its operator norm equals its largest absolute eigenvalue.

- Note that the operator norm in this case is bounded as

$$\begin{aligned}\|(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})\|_2 &= \max \left\{ \left| \lambda_{\min}(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}) \right|, \left| \lambda_{\max}(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X}) \right| \right\} \\ &\leq \underbrace{\max |1 - \eta L|, |1 - \eta m|}_{=: \alpha}.\end{aligned}$$

- Thus we have

$$\begin{aligned}\|\theta^t - \theta^*\|_2 &= \|(\mathbf{I} - \eta \mathbf{X}^\top \mathbf{X})\|_2^t \|\theta^0 - \theta^*\|_2 \\ &\leq \alpha^t \|\theta^0 - \theta^*\|_2\end{aligned}$$

- So as long as

$$\alpha := \max |1 - \eta L|, |1 - \eta m| < 1$$

we have a geometric rate of convergence.

- Verify that this holds for any $\eta \in (0, 2/L)$.

why this holds for any $\eta \in (0, \frac{2}{L})$?

for $\lambda = L$ or m

$$-1 < 1 - \eta \lambda < 1$$

$$m < L$$

↓

L is the strictest constraint

∴ L is steeper slope than m

$$-1 < 1 - \eta L < 1$$

$$-2 < -\eta L < 0$$

$$0 < \eta L < 2$$

$$0 < \eta < \frac{2}{L}$$

Gradient Descent on Least Squares

- That is

$$\|\theta^t - \theta^*\|_2 \leq \epsilon \quad \text{for } t \geq \frac{\log(\|\theta^0 - \theta^*\|_2/\epsilon)}{\log(1/\alpha)}.$$

- Verify that α is guaranteed to lie in between 0 and 1 for any step size that satisfies $\eta \in (0, 2/L)$.

To reach accuracy ϵ , we need:

$$\alpha^t \| \theta^0 - \theta^* \|_2 \leq \epsilon$$

Solve for t (# of iterations).

$$\alpha^t \leq \frac{\epsilon}{\| \theta^0 - \theta^* \|_2}$$

$$t \log(\alpha) \leq \log\left(\frac{\epsilon}{\| \theta^0 - \theta^* \|_2}\right)$$

Since $0 < \alpha < 1$, $\log(\alpha) < 0$

identity:

$$t \geq \frac{\log\left(\frac{\epsilon}{\| \theta^0 - \theta^* \|_2}\right)}{\log(\alpha)}$$

$$\log\left(\frac{a}{b}\right) = -\log\left(\frac{b}{a}\right)$$

$$= \frac{-\log(\| \theta^0 - \theta^* \|_2 / \epsilon)}{-\log(1/\alpha)}$$

$$= \frac{\log(\| \theta^0 - \theta^* \|_2 / \epsilon)}{\log(1/\alpha)}$$